

Question number	Answer	Additional guidance	Mark
6(a)(i)	An explanation that includes the following points: • mutation (in the DNA) causing allele for resistance to the new chemical (1) • (new) chemical acts as a selection pressure (1) • resistant flies survive and pass this {(new) gene / allele} onto their offspring (1)	ACCEPT forming a (new) gene for resistance DO NOT ACCEPT immune for resistant selection pressure causes mutation DO NOT ACCEPT immune for resistant	(3)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	flies have a short life cycle / high mutation rate in flies / flies produce many offspring / chemical is a strong selection pressure	DO NOT ACCEPT selection pressure causes mutation	(1)

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*6(b)	Indicative content: Graph: - black stripes reduce number of biting flies on body and legs (D) 128 down to 111 / by 17 / by 13% (D) - this may not be significant as error bars overlap (D) - black and white stripes reduce number of biting flies on body and legs (D) 128 down to {57/58} / by {70/71} / by 55% (D) - probably significant as error bars do not overlap (D) - black and white stripes reduce the number of biting flies on body and legs more than just black stripes (C) 111 down to {57/58} / by {53/54} / by {48/49}% (C) - probably significant as error bars do not overlap between the two groups of painted cattle (C) - error bar for black stripes is 17% and 17% for black and white stripes (C) Table 1: - black stripes reduce number of biting flies on legs but not on body (D) 1309 down to 1030 / by 279 / by 21% (D) - black and white stripes reduce number of biting flies on body and legs (D) (body) 662 down to 231 / by 431 / by 65% (D) (legs) 1309 down to 710 / by 599 / by 46% (D) - black and white stripes reduce the number of biting flies on body and legs more than just black stripes (C) (body) 677 down to 231 / by 446 / by 66% (C) (legs) 1030 down to 710 / by 320 / by 31% (C) Table 2: - black stripes have no effect on {flicking tail / stamping feet / twitching} (D) - black and white stripes have very little effect on {flicking tail / stamping feet} (D) - black and white stripes have small increase on skin twitching (D) 5 up to 8 / by 3 / by 60% (D)		

	Level 1 1 mark : description of data from one source 2 marks : description of data from two sources Level 2 3 marks : comparison made between black stripes only and black and white stripes, from one source of data 4 marks : two comparisons made between black stripes only and black and white stripes, from two sources of data Level 3 5 marks : two comparisons made between black stripes only and black and white stripes, from two sources of data + level 2 maths calculation OR a comparison of the error bars between black stripes only and black and white stripes 6 marks : two comparisons made between black stripes only and black and white stripes, from two sources of data + level 2 maths calculation AND a comparison of the error bars between black stripes only and black and white stripes		(6)
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